

**SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY**

MICROCONTROLLER & PLC

MODEL QUESTION PAPER-SET 1

Time: 3 Hour

Max.Marks: 75

PART A

- I. Answer all questions in one word or one sentence. Each question carries 1 Mark
(9*1= 9 marks)

1	Write any two features of 8051 microcontroller.	M 1.01	R
2	State the purpose of PSW in 8051	M1.02	R
3	The addressing mode of the instruction MOV A , @R0 is _____	M2.01	U
4	Differentiate the instructions ADD and ADDC.	M2.02	U
5	Write any two disadvantages of PLC	M3.01	R
6	List any two input devices used in a PLC based system.	M3.03	R
7	List any two factors considered for selecting PLC for a typical application.	M3.04	U
8	Horizontal lines in a ladder diagram are called _____	M4.02	R
9	Write the ladder logic diagram of NOT gate.	M4.03	R

PART B

- II. Answer any eight questions from the following, each question carries 3 marks.

(8*3 = 24 marks)

1	State the purpose of DPTR and stack pointer in 8051 microcontroller.	M1.02	U
2	List the registers used for serial communications in 8051 and state their purpose.	M1.04	U
3	State the function of the following pins of 8051 microcontroller. i) RST ii) RXD iii) TXD	M1.04	U
4	Explain the JUMP and CALL instructions in 8051	M2.02	U
5	Write the control word format of 8255 programmable peripheral interface	M2.03	U
6	List any three advantages of PLC	M3.01	R
7	Illustrate the operation of PLC	M3.03	U
8	List any three general features of PLC	M3.01	R
9	With the help of a diagram state the concepts of ladder programming.	M4.02	U
10	Draw the ladder diagrams of (i) AND gate (ii) OR gate	M4.04	U

PART C

Answer ALL questions. Each question carries 7 marks.

(6*7 = 42 marks)

III	Sketch and explain the schematic block diagram of 8051	M1.02	U
	OR		
IV	Explain the alternate functions of port 3 pins of 8051 microcontroller	M1.04	U
V	Illustrate the internal memory organisation of 8051 microcontroller	M1.03	U
	OR		
VI	Sketch the pin configuration of 8051 microcontroller and label all pins	M1.04	U
VII	Explain the rotate and swap instructions of 8051 microcontroller.	M2.02	U
	OR		
VIII	Explain the following addressing modes with examples i) Immediate addressing mode ii) Register addressing mode iii) Direct addressing mode	M2.01	U

IX	Explain the content of register A, Carry flag CY and external memory 8000h after executing the following instructions If the content of external memory location 8000 h is 35 h .Show all necessary steps MOV DPTR, #8000 h MOVXA, @DPTR MOV0F,# 20 h MOV R0, 0Fh ADD A, R0 INC DPTR MOVX A@DPTR, A	M2.02	A
	OR		
X	Write a program to add two 8 bit numbers stored in memory locations 2500H and 2501H and store the result in 2502H and 2503H	M2.02	A
XI	Draw and explain the block diagram of a PLC based system	M3.03	U
	OR		
XII	Compare PLC based panel system with a conventional relay panel system	M3.02	U
XIII	Write a ladder program to realize a starter for interfacing a 1 hp induction motor and list the sequence of events.	M4.04	A
	OR		
XIV	Develop a ladder program to realize staircase light control and explain	M4.04	A

BLUE PRINT

Mark Distribution

Mod ule	H r / M od ul e	(hi / Σ Hi) * 123	TYPE OF QUESTIONS							
			PART A		PART B		PART C		TOTAL	
			No of Question s	Mark s	No of Question s	Marks	No of Question s	Marks	No of Question s	Marks
I	19	32.0 1	2		3		4		9	
				2		9		28		39
II	20	33.7	2		2		4		8	
				2		6		28		36
III	17	28.6 4	3		3		2		8	
				3		9		14		26
IV	17	28.6 4	2		2		2		6	
				2		6		14		22
Total	73	123	9		10		12		31	
				9		30		84		123

Cognitive Level Wise Question analysis

Mark Distribution

Cogn itive Level	% Ma rks	Mar ks	TYPE OF QUESTIONS							
			PART A		PART B		PART C		TOTAL	
			No of Question s	Mark s	No of Question s	Mark s	No of Question s	Marks	No of Question s	Marks
R	30	36.9	6		2		0		8	
				6		6		0		12
U	50	61.5	3		8		8		19	
				3		24		56		83
A	20	24.6	0		0		4		4	
				0		0		28		28
Total	10		9		10		12		31	

1	0	123		9		30		84		123
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Question Wise Analysis

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M 1.01	R	1	2
I.2	M1.02	R	1	2
I.3	M2.01	U	1	2
I.4	M2.02	U	1	2
I.5	M3.01	R	1	2
I.6	M3.03	R	1	2
I.7	M3.04	U	1	2
I.8	M4.02	R	1	2
I.9	M4.03	R	1	2
II.1	M1.02	U	3	7
II.2	M1.04	U	3	7
II.3	M1.04	U	3	7
II.4	M2.02	U	3	7
II.5	M2.03	U	3	8
II.6	M3.01	R	3	7
II.7	M3.03	U	3	8
II.8	M3.01	R	3	7
II.9	M4.02	U	3	8
II.10	M4.04	U	3	7
III.	M1.02	U	7	17
IV.	M1.04	U	7	17
V	M1.03	U	7	17
VI	M1.04	U	7	17
VII	M2.02	U	7	17

VIII	M2.01	U	7	17
IX	M2.02	A	7	17
X	M2.02	A	7	17
XI	M3.03	U	7	17
XII	M3.02	U	7	17
XIII	M4.04	A	7	17
XIV	M4.04	A	7	17
Total			123	295

**SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY**

MICROCONTROLLER & PLC

MODEL QUESTION PAPER-SET 2

Time: 3 Hour

Max.Marks: 75

PART A

- I. Answer **all** questions in one word or one sentence. Each question carries 1 mark.
(9*1=9 marks)

1	_____ is an example of 16 bit register in 8051 microcontroller	M1.02	R
2	List any two applications of 8051 microcontroller.	M1.04	R
3	Match the following instructions and their addressing modes: 1.MOV A, R7 - Indirect addressing mode 2.MOV 80 h, 02 h - Direct addressing mode 3.MOV A, @R0 - Register addressing mode	M2.01	U
4	Write the operation of instructions JNC	M2.02	R
5	Define PLC?	M3.01	R
6	List any two features of PLC	M3.01	R
7	List the classification of PLC Programming Languages	M4.01	R
8	Vertical lines in a ladder diagram are called _____	M4.02	R
9	Write any two Timer instructions in PLC	M4.03	R

PART B

II. Answer any eight questions from the following, each question carries 3 marks.

(8*3=24 marks)

1	State the purpose of following special function registers (i) SBUF (ii) SCON (iii) PCON	M1.02	U
2	List the two registers used for interrupt handling and write the bit format of any one register	M1.02	R
3	Specify the exact addressing mode of the following instructions a) MOV R0, A b) MOV A,@R0 c) MOV A, #65H	M2.01	A
4	Write the bit level operations of 8051 microcontroller	M2.02	R
5	Draw the scan cycle of a typical PLC	M3.03	U
6	Draw the block diagram of PLC	M3.03	U
7	Summarize the applications of PLC	M3.04	U
8	Explain the concept of ladder programming. State the rules for writing a ladder program.	M4.02	R
9.	List any three comparison instructions of PLC	M4.03	R
10	Draw a ladder program to realize XOR gate	M4.04	U

PART C

Answer ALL questions. Each question carries 7 marks.

(6*7= 42 marks)

III	Draw the bit format of PSW register and explain each bit	M1.02	U
	OR		
IV	Sketch the internal architecture of 8051 microcontroller	M1.02	U
V	Compare microprocessor and microcontroller	M1.01	U
	OR		
VI	State the necessity of interrupts in 8085 microcontroller. Also list the interrupt priorities.	M1.02	U

VII	Write a program to find the square of an 8 bit number stored in memory locations 4200H and store the result in 4201H(Least Significant Bytes)and 4202H(Most Significant Bytes)	M2.02	A
	OR		
VIII	Interpret the meaning of the following instructions MOV R0,#20H INC R0 ADD A, R5 MUL AB ANL A, # 15 H XCH A, R0 PUSH B	M2.01	U

IX	Draw and explain the block diagram of 8255 programmable peripheral interface	M2.02	A
	OR		
X	Sketch and explain the interfacing of an 8051 microcontroller with a DC motor.	M2.02	A
XI	Write the criterion for the selection of PLC.	M3.03	U
	OR		
XII	Compare conventional and PLC based control panels	M3.02	U
XIII	Write a ladder program to realize Direct On Line Starter and list the sequence of events.	M4.04	A
	OR		
XIV	Write a ladder program to realize star- delta starter for a three phase induction motor.	M4.04	A

BLUE PRINT

Mark Distribution

Module	Hr / Module	(hi / $\sum H_i$) * 123	TYPE OF QUESTIONS							
			PART A		PART B		PART C		TOTAL	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
I	19	32.01	2	2	2	6	4	28	8	36
II	20	33.7	2	2	2	6	4	28	8	36
III	17	28.64	2	2	3	9	2	14	7	25
IV	17		3		3		2		8	

		28.6 4		3		9		14		26
Tot al	73	123	9	9	10	30	12	84	31	123

Cognitive Level Wise Question Analysis

Mark Distribution

Cog nitiv e Leve l	% Ma rks	Ma rks	TYPE OF QUESTIONS							
			PART A		PART B		PART C		TOTAL	
			No of Questio ns	Marks	No of Questio ns	Marks	No of Questio ns	Marks	No of Questio ns	Marks
R	30	36. 9	8		4		0		12	
				8		12		0		20
U	50	61. 5	1		5		8		14	
				1		15		56		72
A	20	24. 6	0		1		4		5	
				0		3		28		31
Tot al	100	123	9	9	10	30	12	84	31	123

Question Wise Analysis

Course :	MICROCONTROLLER & PLC			
Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.02	R	1	2
I.2	M1.04	R	1	2
I.3	M2.01	U	1	2
I.4	M2.02	R	1	2
I.5	M3.01	R	1	2
I.6	M3.01	R	1	2
I.7	M4.01	R	1	2
I.8	M4.02	R	1	2
I.9	M4.03	R	1	2
II.1	M1.02	U	3	7
II.2	M1.02	R	3	7
II.3	M2.01	A	3	7
II.4	M2.02	R	3	7
II.5	M3.03	U	3	8
II.6	M3.03	U	3	8
II.7	M3.04	U	3	7
II.8	M4.02	R	3	8
II.9	M4.03	R	3	7
II.10	M4.04	U	3	7
III.	M1.02	U	7	17
IV.	M1.02	U	7	17
V	M1.01	U	7	17
VI	M1.02	U	7	17
VII	M2.02	A	7	17

VIII	M2.02	A	7	17
IX	M2.03	U	7	17
X	M2.04	U	7	17
XI	M3.04	U	7	17
XII	M3.02	U	7	17
XIII	M4.02	A	7	17
XIV	M4.04	A	7	17
Total			123	295

Scoring Indicators

Model Question Paper I

MICROCONTROLLER & PLC

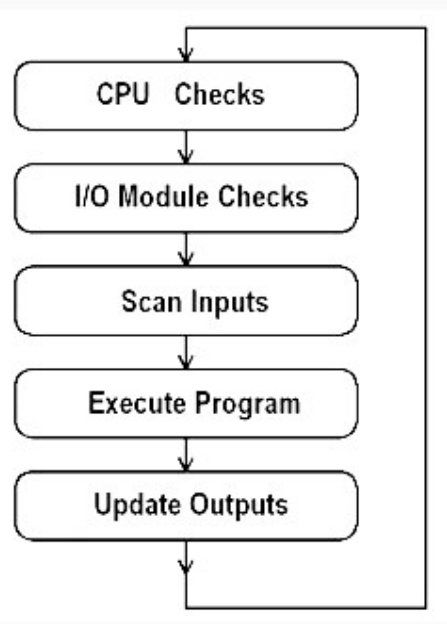
Q No	Scoring Indicators	Split score	Sub Total	Total Score
	PART A			
I. 1	<u>Features of 8051 MC</u> • 8 bit ALU • 4KB bytes on-chip program memory (ROM) • 128 bytes on-chip data memory (RAM) • Four register banks. • 8-bit bidirectional data bus. • 16-bit unidirectional address bus. • 8 bit ports - 4 nos • 32 general purpose registers each of 8-bit. • 16 bit Timers (2 Nos) • Serial port - 1 No (full duplex) <i>(write any two)</i>	2*0.5	1	1
I. 2	The Program Status Word (PSW) contains status bits that reflect the current CPU state.	1	1	1
I. 3	Indirect addressing mode	1	1	1
I. 4	ADD :- Will perform a simple addition function. ADDC :- Will perform addition function with carry.	2*0.5	1	1
I. 5	<u>Disadvantages of PLC</u> 1. There's too much work required in connecting wires. 2. There's difficulty with changes or replacements. 3. It's always difficult to find errors 4. It requires a skillful workforce. 5. When a problem occurs, hold-up time is indefinite, usually long. <i>(write any two)</i>	2*0.5	1	1

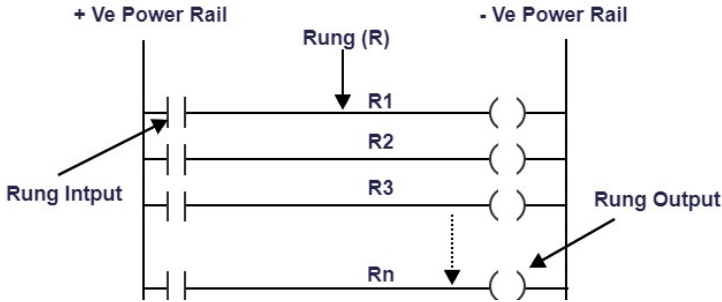
I. 6	DIP switch Push button Reed switch Limit switch Light sensors Proximity switch <i>(write any two)</i>	2*0.5	1	1
I. 7	• Number of I/O's: • Type of I/O's • Memory and Programming Requirements • Compact or Modular PLC • Instruction Set/CPU • PLC Scan Time • Sinking & Sourcing PLC • Manufacturer's Support and Backup <i>(write any two)</i>	2*0.5	1	1
I. 8	Horizontal lines in a ladder diagram are called “rungs,”	1	1	1
I. 9	 <i>(or similar)</i>	1	1	1
	PART B			

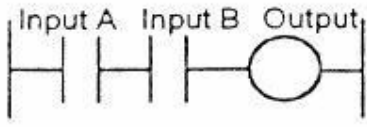
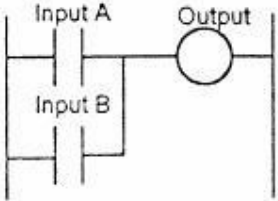
II. 1	<u>Stack pointer and DPTR</u> <i>Stack pointer- 1.5</i> <i>DPTR- 1.5</i> Stack Pointer: In the 8051, the stack pointer (SP) points to the top of the stack .(The stack refers to an area of internal RAM to store and retrieve data quickly.Stack pointer holds this internal RAM address that is called top of the stack) DPTR: It is used by the 8051 to access external memory using the address indicated by DPTR. DPTR is the only 16-bit register available for this purpose and is often used to store 2-byte values.	2*1.5	3	3
II. 2	<u>Serial communication registers</u> SBUF (Serial Buffer Register)- to store 8 bit data SCON (Serial Control Register)-used for mode selection PCON (Power Control Register)- used for selection of baud rate	3*1	3	3
II. 3	i)RST- It is a RESET pin, which is used to reset the microcontroller to its initial values. ii)RXD (serial data receive pin) which is for serial input. Through this input signal microcontroller receives data for serial communication. iii)TXD: (serial data transmit pin) which is serial output pin. Through this output signal microcontroller transmits data for serial communication. <i>(Each pin carries 1 mark)</i>	3*1	3	3

II. 4	JUMP Instruction • By using JUMP, the program control transfers to a location which is also a part of the main program • Here the addressing mode is ImmediateHere the addressing mode is Immediate • The program counter value is not Pushed into Stack CALL Instruction • By using CALL instruction, the program control transfers to a location which is not a part of main program • Here the addressing mode is Immediate and Register Indirect. • The value of the program counter is pushed into the stack before going to the pointed location. <i>(each carries 1.5 mark)</i>	2*1.5	3	3
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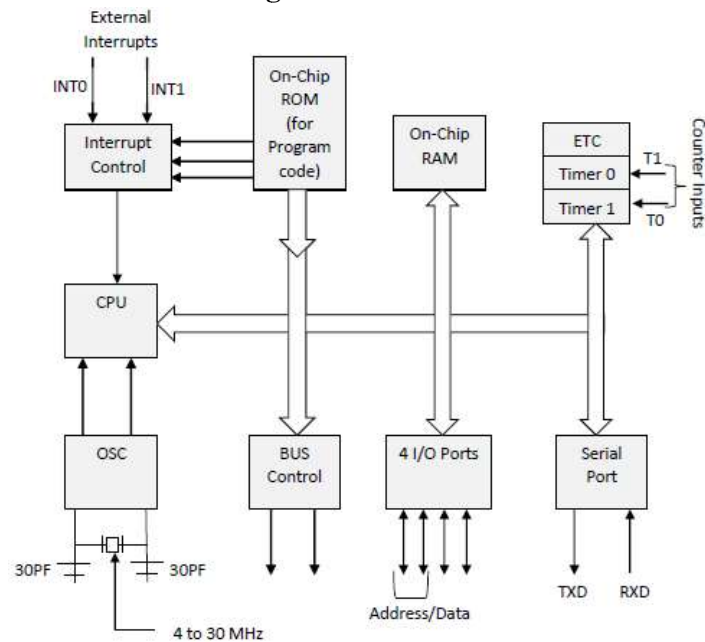
II. 5	<u>Control Word of 8255 programmable peripheral interface</u> The diagram illustrates the bit fields of the 8255 Control Word: Group B: - Port C (Lower—PC₇–PC₀): 1 = Input, 0 = Output - Port B: 1 = Input, 0 = Output - Mode Selection: 0 = Mode 0, 1 = Mode 1 Group A: - Port C (Upper—PC₇–PC₄): 1 = Input, 0 = Output - Port A: 1 = Input, 0 = Output - Mode Selection: 00 = Mode 0, 01 = Mode 1, 1X = Mode 2 Global Mode: 1 = I/O Mode, 0 = BSR Mode	3	3	3
II 6.	<u>Advantages of PLC</u> 1. Reliability: The internal relay systems of a PLC are solid state. This means that the relay function is not mechanical like conventional relay systems and components. 2. Ease of Troubleshooting: Another advantage of a PLC system is the ease of troubleshooting. In a PLC system, a good technician can read through the programming and usually figure out what is and isn't working. 3. Easy Expandability: One of the best features that a PLC system has over a traditional relay system is versatility with the programming and easy expandability. 4. Smaller Size: The physical size requirements of a PLC system are far smaller than a cabinet needed for relay logic circuitry. 5. PLC can support a wide variety of field components 6. They have significant advantages over traditional control systems based on relay <i>(Write any three)</i>	1*3	3	

II. 7	<u>Operation of PLC</u> PLCs operate by continually scanning programs and repeat this process many times per second. When a PLC starts, it runs checks on the hardware and software for faults, also called a self-test. If there are no problems, then the PLC will start the scan cycle. The scan cycle consists of three steps: input scan, executing program(s), and output scan. Input Scan : PLC reads the status of the inputs and solves the logic. Execute Program (or Logic Execution) : The PLC executes a program one instruction at a time Output Scan : When the ladder scan completes, the PLC updates the status of the outputs  <pre> graph TD A[CPU Checks] --> B[I/O Module Checks] B --> C[Scan Inputs] C --> D[Execute Program] D --> E[Update Outputs] E --> A </pre> (Explanation or flow chart - any one)	3	3	3
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II. 8	List general features of PLC (any three Each carry 1 mark) ● Processor Module: Single microprocessor device ● Architecture: Programmable Logic Controllers (PLC) was designed to replace the relay-based systems ● Programmable Logic Controllers are generally Specified by ○ No of inputs available ○ No of output available ○ Types of input available - analog/digital ○ types of power supply -ac/dc ○ programming language ● Programming language -There are programming languages for PLC including ladder diagram, SFC, STL, ST and so on. ● Other general features ○ High reliability. ○ Strong anti-interference quality ○ Good flexibility. ○ Easy-Operation <i>(Write any three features)</i>	3*1	3	3
II.9	Ladder logic is a programming language that is used to program a PLC (Programmable Logic Controller). It is a graphical PLC programming language which expresses logic operations with symbolic notation using ladder diagrams, much like the rails and rungs of a traditional relay logic circuit.  <i>Explanation: 1.5</i> <i>Diagram: 1.5</i>	2*1.5	3	3

II.10	Input A Input B Output  AND Input A Output  OR <i>(Each carries 1.5 marks)</i>	2*1.5	3	3
	PART C			

Schematic block diagram of 8051



Functions of each block :

CPU (Central Processing Unit): CPU acts as a mind of any processing machine. It synchronizes and manages all processes that are carried out in microcontrollers.

Memory: In microcontroller 8051 there is 4 KB ROM and data memory (RAM) of 128 bytes.

Bus : Bus is a group of wires which uses as a communication canal or acts as means of data transfer.

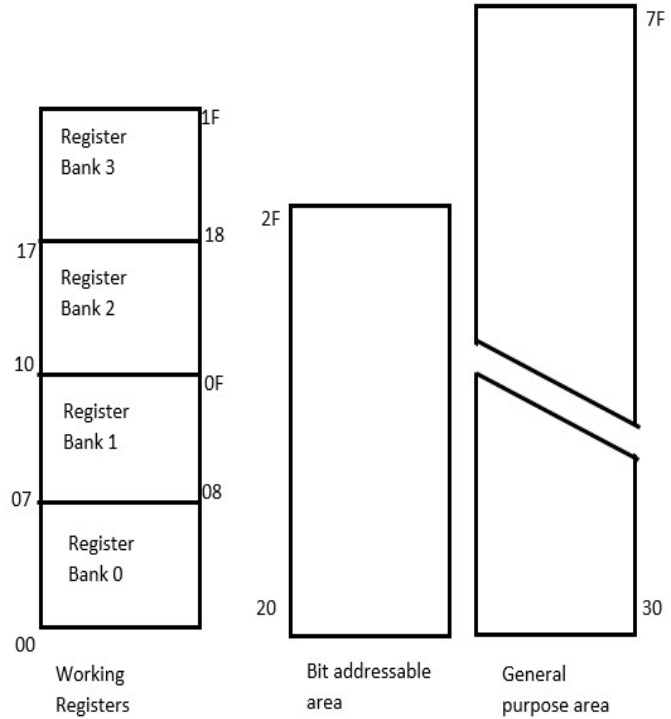
Oscillator: As the microcontroller is a digital circuit therefore it needs timer for their operation. To perform timer operation inside microcontroller it required externally connected or on-chip oscillator

Interrupt control: To control all the interrupt handling process

Sketch - 4

Explanation - 3

IV	<u>Alternate functions of port 3 pins of 8051 microcontroller</u>	7*1	7																												
<table><tr><td>P 3 bit</td><td>function</td><td>Description</td></tr><tr><td>P3.0</td><td>RXD</td><td>Serial Data Reception</td></tr><tr><td>P3.1</td><td>TXD</td><td>Serial Data Transmission</td></tr><tr><td>P3.2</td><td><u>□□□□</u></td><td>External Interrupt 0</td></tr><tr><td>P3.3</td><td><u>□□□I</u></td><td>External Interrupt 1</td></tr><tr><td>P3.4</td><td>TO</td><td>Timer 0</td></tr><tr><td>P3.5</td><td>T1</td><td>Timer 1</td></tr><tr><td>P3.6</td><td><u>□□</u></td><td>Write Signal</td></tr><tr><td>P3.7</td><td><u>□□</u></td><td>Read Signal</td></tr></table>					P 3 bit	function	Description	P3.0	RXD	Serial Data Reception	P3.1	TXD	Serial Data Transmission	P3.2	<u>□□□□</u>	External Interrupt 0	P3.3	<u>□□□I</u>	External Interrupt 1	P3.4	TO	Timer 0	P3.5	T1	Timer 1	P3.6	<u>□□</u>	Write Signal	P3.7	<u>□□</u>	Read Signal
P 3 bit	function	Description																													
P3.0	RXD	Serial Data Reception																													
P3.1	TXD	Serial Data Transmission																													
P3.2	<u>□□□□</u>	External Interrupt 0																													
P3.3	<u>□□□I</u>	External Interrupt 1																													
P3.4	TO	Timer 0																													
P3.5	T1	Timer 1																													
P3.6	<u>□□</u>	Write Signal																													
P3.7	<u>□□</u>	Read Signal																													
<i>(Each carries 1 mark)</i>																															

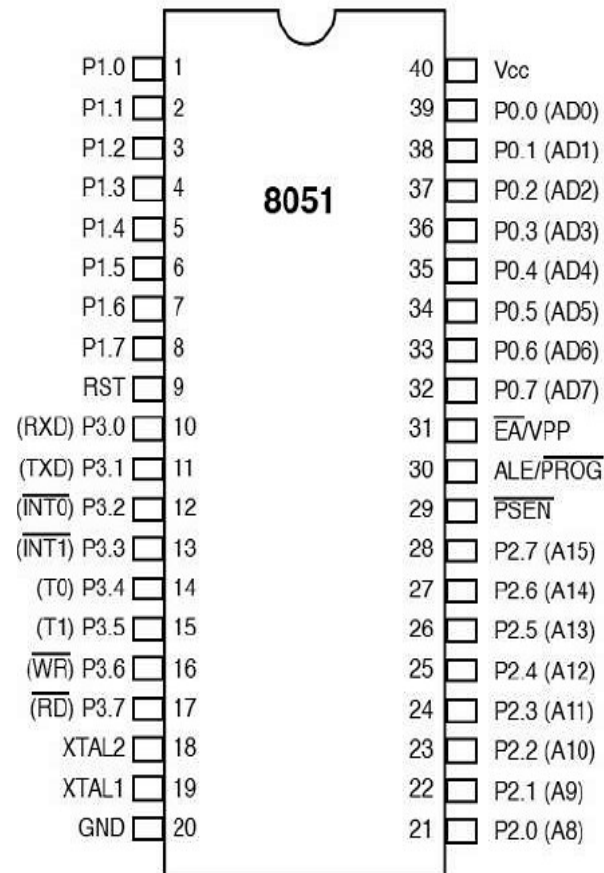
V	<u>Internal memory organisation of 8051 microcontroller</u> Internal Data Memory (RAM) The Data Memory or RAM of the 8051 Microcontroller stores temporary data and intermediate results that are generated and used during the normal operation of the microcontroller. the 8051 Microcontroller have 128B of RAM. i.e., memory addresses from 00H to 7FH and are divided into Working Registers (organized as Register Banks), Bit – Addressable Area and General Purpose RAM In the first 128B of RAM (from 00H to 7FH), the first 32B i.e., memory from addresses 00H to 1FH consists of 32 Working Registers that are organized as four banks with 8 Registers in each Bank.  The diagram illustrates the internal memory organization of the 8051 microcontroller. It is divided into three main sections: Working Registers, Bit addressable area, and General purpose area. The Working Registers section (00H to 1FH) is further divided into four banks: Register Bank 0 (00H-07H), Register Bank 1 (08H-0FH), Register Bank 2 (10H-17H), and Register Bank 3 (18H-1FH). The Bit addressable area (20H-2FH) is a 16B area. The General purpose area (30H-7FH) is a 48B area. Internal programme memory (ROM) Program Memory (ROM) is used for a permanent saving	3+4	7	
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VI

Pin configuration of 8051 microcontroller*sketch- 3 mark**Labelling- 4 mark*

7

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VII	<u>Rotate and Swap instructions</u> Register A Rotate Operations <i>Rotate instruction- 4 * 1.5 mark</i> <i>Swap-1 mark</i>	6+1	7	
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VIII	<u>Addressing modes</u> (Immediate addressing mode : 2mark Register addressing mode: 2 mark Direct addressing mode: 3 mark) Immediate addressing mode: In this addressing mode, the data is directly specified in the instruction itself. The source is the immediate data and the destination of this instruction could be any register. MOV A, #65H ; The digit 65H is copied to the Accumulator (A). MOV DPTR, #1615H ; A 16 bit digit 1615H is copied to a Data pointer which is 16 bits. Register addressing mode: In register addressing mode, the source and the destination both are registers, and must be of the same size. Only the Accumulator (A) and R0 to R7 registers of each memory bank are allowed in this mode to transfer the data. The data transfer can take place between Rn(R0 to R7) registers and the Accumulator (A) only and cannot be done between Rn registers. Ex: MOV A, R3 ; The contents of register R3 are copied to Accumulator (A). ADD A, R4 ; The contents of register R4 are added with the contents of the Accumulator (A) and the result is stored in the Accumulator (A). Direct addressing mode: In this addressing mode, source and destination could be a register or a RAM location, but both cannot be the same, either the source has to be a register followed by RAM location as destination and vice versa. The address of the operand is specified in the instruction itself. Ex: MOV R3, 70H ; The contents of RAM location 70H are copied to the register R3.	3+2+ 2	7	
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IX	Explain the content of register A, Carry flag CY and external memory 8001 H after executing the following instruction If the content of external memory location 8000 h is 35 h . Show all necessary steps	1*7	7	7														
<table><tr><td>MOV DPTR, #8000 h</td><td>[DPTR] ←---8000 h</td></tr><tr><td>MOVXA, @DPTR</td><td>[A] ←-----EXT[(DPTR)] [A] ←-----EXT[8000] [A] ←-----35 h</td></tr><tr><td>MOV0F,# 20 h</td><td>[0F] ←--- 20 h</td></tr><tr><td>MOV R0, 0Fh</td><td>[R0] ←-----[0F] [R0] ←----- 20 h</td></tr><tr><td>ADD A, R0</td><td>[A] ←----- [A] + [R0] [A] ←----- 35 + 20 [A] ←----- 55 h, [CY]= 0</td></tr><tr><td>INC DPTR</td><td>[DPTR] ←--- [DPTR] + 1 [DPTR] ←--- 8000+ 1 DPTR] ←--- 8001</td></tr><tr><td>MOVX A@DPTR, A</td><td>EXT [(DPTR)] ←-----[A] EXT[8001] ←-----55</td></tr></table>		MOV DPTR, #8000 h	[DPTR] ←---8000 h	MOVXA, @DPTR	[A] ←-----EXT[(DPTR)] [A] ←-----EXT[8000] [A] ←-----35 h	MOV0F,# 20 h	[0F] ←--- 20 h	MOV R0, 0Fh	[R0] ←-----[0F] [R0] ←----- 20 h	ADD A, R0	[A] ←----- [A] + [R0] [A] ←----- 35 + 20 [A] ←----- 55 h, [CY]= 0	INC DPTR	[DPTR] ←--- [DPTR] + 1 [DPTR] ←--- 8000+ 1 DPTR] ←--- 8001	MOVX A@DPTR, A	EXT [(DPTR)] ←-----[A] EXT[8001] ←-----55			
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MOVX A@DPTR, A	EXT [(DPTR)] ←-----[A] EXT[8001] ←-----55																	
(Each step carries 1 mark)																		

X	Write a program to add two 8 bit numbers stored in memory location 2500H and 2501H and store the result in 2502H and 2503H (program- 7marks) <pre> 3500 ORG 3500H MOV DPTR, #2500 h MOVX A, @DPTR ; First number in A MOV R0, A INC DPTR MOVX A, @DPTR : second number in A MOV R7, #00H : To accommodate carry ADD A,R0 JNC LOOP INC R7 : set carry =1 LOOP: INC DPTR MOVX @DPTR, A MOV A, R7 INC DPTR MOVX @DPTR, A SJMP:HALT END </pre> <i>or any similar program</i> <i>(for algorithm or flowchart without program -3 mark)</i>	7		7
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XI	<u>Block diagram of PLC</u> <i>Block diagram : 4</i> <i>Explanation : 3</i> <pre> graph LR subgraph PLC_Box [] direction LR IM[Input Module] --> CPU[Central Processing Unit CPU] CPU --> OM[Output Module] end PS[Power Supply] --> CPU PD[Programming Device] <--> CPU IDS[Input Device Digital / Analog Input Signal] --> IM OM --> ODS[Output Device Digital / Analog Output Signal] </pre> Processor section (CPU): The processor section is the brain of PLC which consists of RAM, ROM, logic solver and user memory. The central processing unit is the heart of PLC. Input and output module: The input module is a mediator between input devices and central processing unit (CPU) which is used to convert analog signal into digital signal. The output module is a mediator between output devices and the central processing unit (CPU) which converts digital signal into analog signal. Power supply: power supply is provided to the processor unit, input and output module unit. Power supply may be an integral or separately mounted unit. Memory section: The memory section is the area of the CPU in which data and information is stored and retrieved. Programming device: Programming devices are dedicated devices used for loading the user program into the program memory or edit it and to monitor the execution of the program of the PLC	4+3	7	
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XII	Comparison between PLC based panel and a conventional relay panel system																											
	<table><tr><td></td><td>PLC</td><td>Relay</td></tr><tr><td>1)Basic</td><td>It is a solid-state computerized industrial controller that performs software logic by using input & output modules, CPU, memory, and others.</td><td>Relay is an electro-mechanical switching hardware device (Hardware Switching Device)</td></tr><tr><td>2)Function</td><td>It plays a monitoring as well as controlling role in designing circuits.</td><td>It plays only a controlling role in the designing circuit. Monitoring is not so easy with a relay.</td></tr><tr><td>3) Working:</td><td>In PLC, we can write the program using different types of programming languages.</td><td>In Relay, we cannot write the program.</td></tr><tr><td>4) Design:</td><td>Easily modify the designing circuit.</td><td>Modification of the electronic circuit is more difficult as compared to PLC</td></tr><tr><td>5)Memory</td><td>consist of memory to store programme</td><td>no memory</td></tr><tr><td>6) I/O</td><td>It has more capabilities of input and output modules.</td><td>The relay does not have more capabilities.</td></tr><tr><td>7) Operation:</td><td>PLC is operated on the digital system.</td><td>It is operated on the analog system.</td></tr></table>		PLC	Relay	1)Basic	It is a solid-state computerized industrial controller that performs software logic by using input & output modules, CPU, memory, and others.	Relay is an electro-mechanical switching hardware device (Hardware Switching Device)	2)Function	It plays a monitoring as well as controlling role in designing circuits.	It plays only a controlling role in the designing circuit. Monitoring is not so easy with a relay.	3) Working:	In PLC, we can write the program using different types of programming languages.	In Relay, we cannot write the program.	4) Design:	Easily modify the designing circuit.	Modification of the electronic circuit is more difficult as compared to PLC	5)Memory	consist of memory to store programme	no memory	6) I/O	It has more capabilities of input and output modules.	The relay does not have more capabilities.	7) Operation:	PLC is operated on the digital system.	It is operated on the analog system.	7*1	7	
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(any 7 comparisons- each carries 1 mark)																												

XIII	Inputs: START- To start the motor- NO STOP - To stop the motor-NC OLR- Overload Relay Output: MOTOR- Motor Sequence of Events : 1. When the Start push button is pressed, Motor has to start. 2. If the Start is released and the Stop pushbutton is not pressed, Motor should remain on. 3. When the Stop push button is pressed, Motor has to stop. 4. If stop push button is released and start is not pressed (released) motor should remain off. 5. When an overload occurs the motor stops <i>Program - 5</i> <i>Explanation - 2</i> <i>(or similar ladder program)</i>	5+2	7	
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XIV

Staircase lighting control

The diagram is a ladder logic circuit for a staircase lighting control system. It consists of two parallel rungs connected to a common rail on the right. The top rung contains two normally open contacts labeled 'I1' and 'I2' in series, with 'SWITCH 1' and 'SWITCH 2' written below them respectively. The bottom rung also contains two normally open contacts labeled 'I1' and 'I2' in series, with 'SWITCH 1' and 'SWITCH 2' written below them respectively. Both rungs lead to a coil labeled 'Q1' and 'LAMP'.

Lamp conditions :

ON	OFF	ON
ON	ON	OFF
OFF	ON	ON
OFF	OFF	OFF

Program-5Mark
Explanation-2 Mark

(or similar ladder logic diagram)

5+2

7

Scoring Indicators

Model Question Paper II

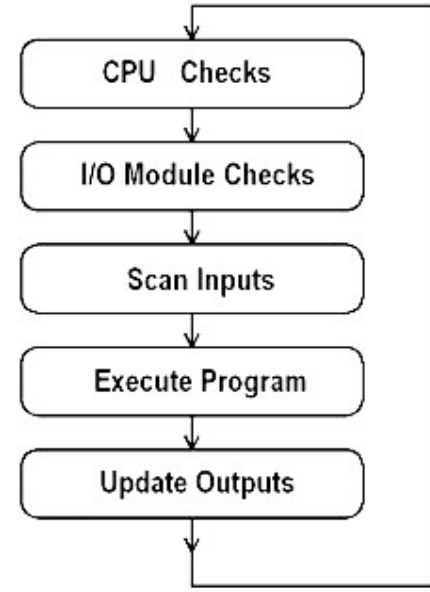
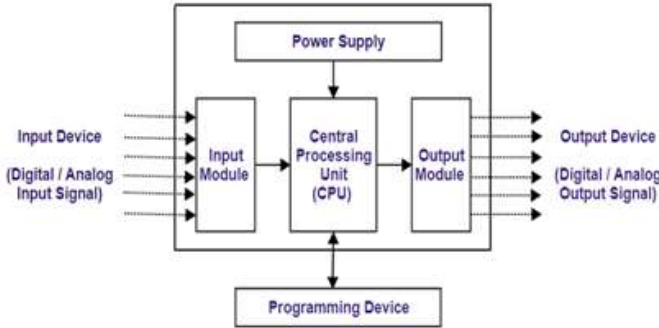
MICROCONTROLLERS & PLC

Q No	Scoring Indicators	Split score	Sub Total	Total score
	PART A			
I. 1	• DPTR • PC • TIMER 0 (T0) • TIMER 1 (T1) (write any one)	1	1	1
I. 2	The use of microcontroller in various fields such as automobile, aeronautics, space, robotics, electronics, defense application, mobile communications, rail transport, industrial processing, and medical applications (write any two)	2*0.5	1	1
I. 3	Match the following instructions and their addressing modes: 1.MOV A, R7 - Register addressing mode 2.MOV 80 h, 02 h - Direct addressing mode 3.MOV A, @R0 - Indirect addressing mode	1	1	1
I. 4	The JNC instruction transfers program control to the specified address if the carry flag is 0.	1	1	1
I. 5	A programmable logic controller (PLC) is an industrial computer control system that continuously monitors the state of input devices and makes decisions based upon a custom program to control the state of output devices	1	1	1

I. 6	● Programmable Logic Controllers are generally Specified by ○ No of inputs available ○ No of output available ○ Types of input available - analog/digital ○ types of power supply -ac/dc ○ programming language ● Other general features ○ High reliability. ○ Strong anti-interference quality ○ Good flexibility. ○ Easy-Operation <i>(write any two)</i>	2*0.5	1	1
I.7	● Ladder Diagram (LD) ● Sequential Function Charts (SFC) ● Function Block Diagram (FBD) ● Structured Text (ST) ● Instruction List (IL) <i>(write any two)</i>	2*0.5	1	1
I. 8	Power rails	1	1	1
I. 9	Three timers are, 1. On Timer (Ton): An on-delay timer (TON) is a programming instruction which use to start momentary pulses for a set period of time. 2. Off Timer (T off): An off-delay (TOF) timer is a PLC programming instruction which use to switch off the output or system after a certain amount of time. 3. Retentive On-Off Timer (RTO): The main function of the RTO is used to hold or store the set (accumulated) time.RTO is used in the case when there is a change in the rung state, power loss, or any interruption in the system. <i>(any two-name only)</i>	2*0.5	1	1
	PART B			

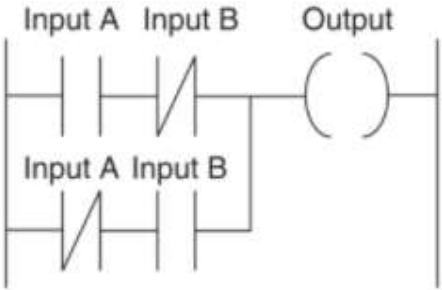
II. 1	<u>Special function registers</u> i) SBUF: The special function register SBUF is physically two registers. One is write-only and is used to hold data to be transmitted out of the 8051 via TXD. The other is, read-only and holds the received data from external sources via RXD. ii) SCON: This register contain the mode selection bits and 9th data bit for transmit and receive (TB8 and RB8) and the serial part interrupt bits (TI and RI) iii) PCON:Power control register-used to control the Baud Rate of the Serial communication. <i>(each carry one mark)</i>	3*1	3	3
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II. 2	InterruptEnable Register(IE) Interrupt Priority Register (IP) Interrupt Enable (IE) Register:. 76543210 <table><tr><td>EA</td><td>X</td><td>X</td><td>ES</td><td>ET1</td><td>EX1</td><td>ET0</td><td>EX0</td></tr></table> ❖ EX0- Enable/Disable – External Interrupt 0 ❖ ET0- Enable/Disable – Timer 0 overflow Interrupt ❖ EX1- Enable/Disable – External Interrupt 1 ❖ ET1- Enable/Disable – Timer 1 overflow Interrupt ❖ ES -Enable/Disable – Serial port Interrupt ❖ EA - Enable/Disable – All interrupts Interrupt Priority Register (IP): <table><tr><td>---</td><td>---</td><td>PT2</td><td>PS</td><td>PT1</td><td>PX1</td><td>PT0</td><td>PX0</td></tr></table> IP.7: reserved IP.6: reserved IP.5: Timer 2 interrupt priority bit (8052 only) IP.4: Serial port interrupt priority bit IP.3: Timer 1 interrupt priority bit IP.2: External interrupt 1 priority bit IP.1: Timer 0 interrupt priority bit IP.0: External interrupt 0 priority bit <i>Listing- 1 mark any one format-2 mark</i>	EA	X	X	ES	ET1	EX1	ET0	EX0	---	---	PT2	PS	PT1	PX1	PT0	PX0	1+2	3	3
EA	X	X	ES	ET1	EX1	ET0	EX0													
---	---	PT2	PS	PT1	PX1	PT0	PX0													
II. 3	a) MOV R0, A -Register addressing mode b) MOV A, @R0 -Indirect addressing mode c) MOV A, #65H- Immediate addressing mode (Each carries one mark)	3*1	3	3																

II. 4	<u>Bit level operations</u> ANL ORL CPL CLR MOV SETB	6*0.5	3	3
II. 5	 <pre> graph TD A[CPU Checks] --> B[I/O Module Checks] B --> C[Scan Inputs] C --> D[Execute Program] D --> E[Update Outputs] E --> A </pre>	3	3	3
II. 6	 (or any similar block diagram)	3*1	3	3

II. 7	1.Industrial Applications of PLC ● The Transportation System likes Conveyor Belt System. ● Packing and Labeling System in Food & Beverage. ● Automatic Bottle or Liquid Filling System. ● Packaging and Labelling System in Pharma Industries. 2) Power Station Applications of PLC ● PLC uses the Smart Grid System to Monitor and Detect fault conditions. ● It is used in the Power Generation, Transmission, and Distribution System. 3)Commercial Applications of PLC: ● Smart Traffic Control Signal System. ● Smart Elevator Control System. ● Fire Detection and Alarm System. 4) Domestic Applications of PLC ● Water Tank Level Control System ● Car Washing and Parking System. ● Flashing Light Controlling System. ● Automatic Door Opening/Closing System. (any three applications)	3*1	3	3
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II.8	Ladder program <i>basic concept: 1</i> <i>any two rules: 2</i> Ladder logic is a programming language that is used to program a PLC (Programmable Logic Controller). It is a graphical PLC programming language which expresses logic operations with symbolic notation using ladder diagrams, much like the rails and rungs of a traditional relay logic circuit. Rules: • Inputs can be used in Series as well as Parallel to form a connection • Outputs (or coil) can be used only in Parallel • One Input can be used in multiple times in one program • One Output cannot be used multiple times in one program, except in Set/Reset and Latch/ Unlatch functions • Input Address cannot be used as an Output Address • Outputs Address can be used as Inputs Address	1+2	3	3
II.9	Comparison instructions: • EQU (equal) • NEQ (not equal) • LEQ (Less than or equal) • GRT (Greater than) • GEQ (Greater than or equal) <i>(write any two)</i>	2*1.5	3	3

II. 10		3	3	3
	PART C			

III

Structure of PSW

PSW Program Status Word Register

7

6

5

4

3

2

1

0

CY

AC

F0

RS1

RS0

OV

—

P

The program status word (PSW) register is an 8-bit register. It is also referred to as the flag register.

CY: Carry flag. This flag is set whenever there is a carry out from the D7 bit after an 8 bit addition or subtraction.

AC: Auxiliary carry flag If there is a carry from D3 and D4 during an ADD or SUB operation, this bit is set;

F0: Available to the user for general purposes.

RS0, RS1: Register bank selects bits These two bits are used to select one of the four register banks from internal RAM

RS1

RS0

Reg Bank Selected

0

0

Bank 0

0

1

Bank 1

1

0

Bank 2

1

1

Bank 3

OV: Overflow flag: This flag is set whenever the result of a signed number operation is too large, causing the high- order bit to overflow into the sign bit.

P: Parity flag The parity flag reflects the number of 1s in the A (accumulator) register only. If the A register contains an odd number of 1s, then P=1. P=0 if A has an even number of 1s.

structure - 3

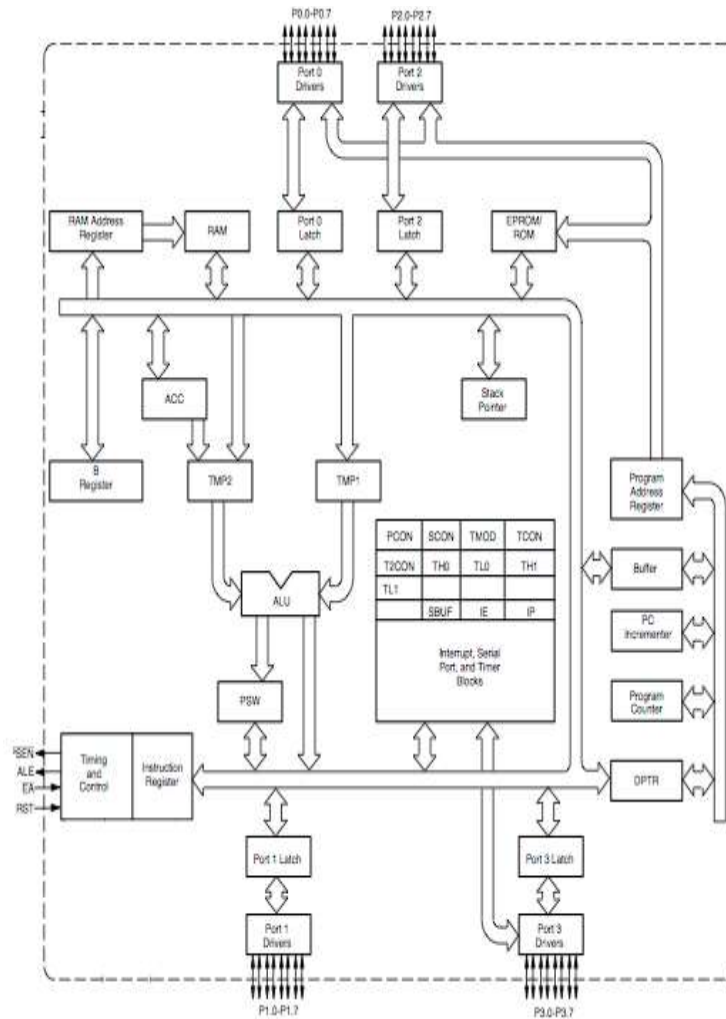
explanation - 4

3+4

7

7

IV

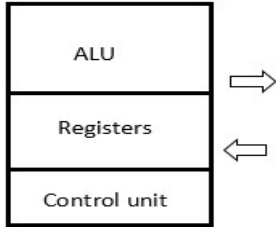
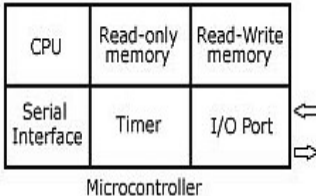
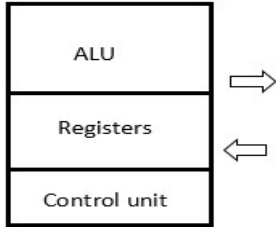
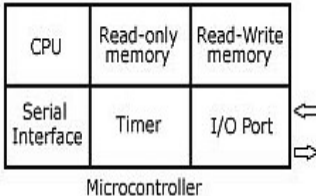
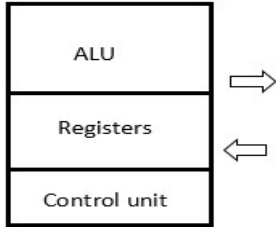
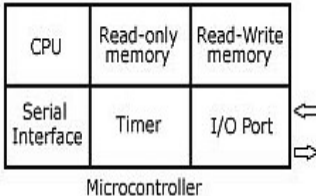


7

7

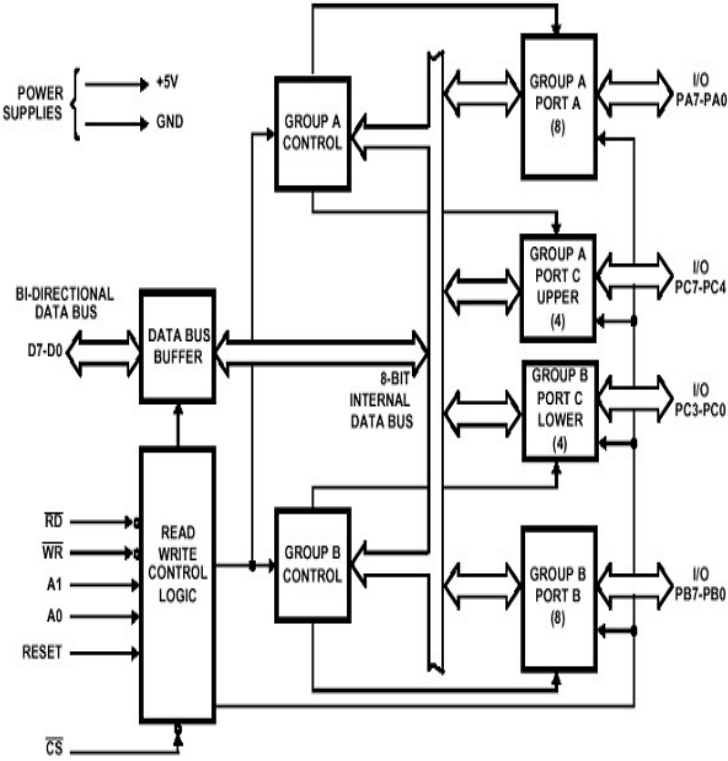
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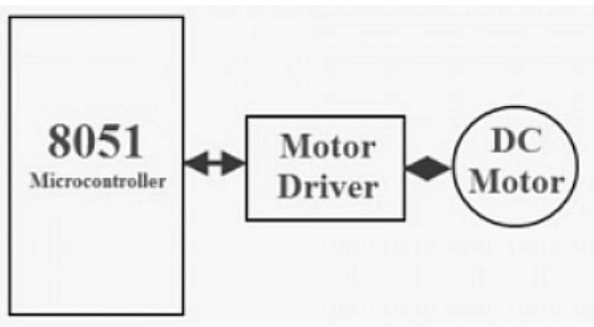
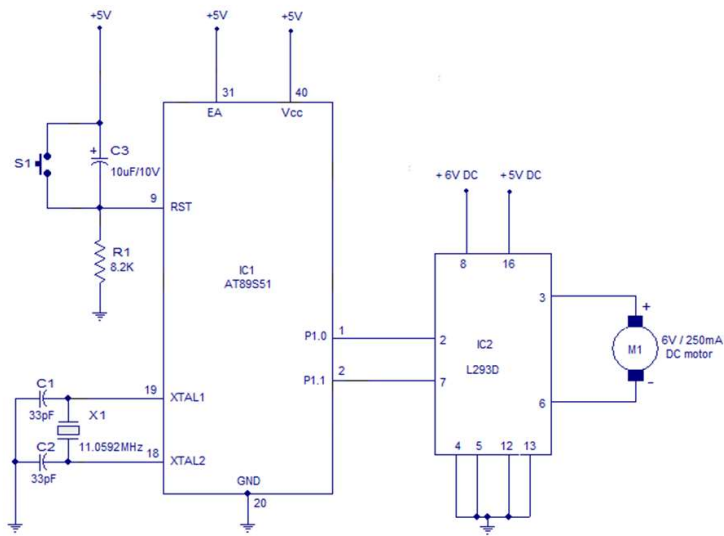
**Sketch (incorporate
major blocks) - 7
or similar diagrams**

V	Compare microprocessor and microcontroller (any 7 points)	7*1	7	7																										
<table><tr><th>Microprocessor</th><th>Microcontroller</th></tr><tr><td></td><td> Microcontroller</td></tr><tr><td>A general Purpose digital computer CPU</td><td>A micro computer implemented on a single chip</td></tr><tr><td>Memory,I/O ports , Timers , Interrupts etc are not available inside the chip</td><td>All are available inside the chip</td></tr><tr><td>Must have many additional digital parts to operational as a system</td><td>Can function as a computer without any additional digital parts</td></tr><tr><td>Make the system bulkier and expensive</td><td>Make the system handy and economic</td></tr><tr><td>Many instructions for moving data between memory and CPU</td><td>Have one or two for this purpose</td></tr><tr><td>Has one or two bit handling instructions</td><td>Many bit handling instructions</td></tr><tr><td>Not capable for bit level boolean functions</td><td>Many bit handling instructions</td></tr><tr><td>Access time is more</td><td>Access time is less</td></tr><tr><td>Very few pins are programmable</td><td>Most of the pins are programmable</td></tr><tr><td>Mostly used in information processing</td><td>Mostly used in control oriented activities</td></tr><tr><td>Based on Von Neumann model</td><td>Based on Harward model</td></tr></table>					Microprocessor	Microcontroller		 Microcontroller	A general Purpose digital computer CPU	A micro computer implemented on a single chip	Memory,I/O ports , Timers , Interrupts etc are not available inside the chip	All are available inside the chip	Must have many additional digital parts to operational as a system	Can function as a computer without any additional digital parts	Make the system bulkier and expensive	Make the system handy and economic	Many instructions for moving data between memory and CPU	Have one or two for this purpose	Has one or two bit handling instructions	Many bit handling instructions	Not capable for bit level boolean functions	Many bit handling instructions	Access time is more	Access time is less	Very few pins are programmable	Most of the pins are programmable	Mostly used in information processing	Mostly used in control oriented activities	Based on Von Neumann model	Based on Harward model
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Based on Von Neumann model	Based on Harward model																													

VI	<u>Interrupts in 8085 microcontroller.</u> (Necessity : 2 5 interrupts and its priorities: 5) Interrupts are the events that temporarily suspend the main program, pass the control to the external sources and execute their task. It then passes the control to the main program where it had left off. Interrupts are important because they give the user better control over the computer. Without interrupts, a user may have to wait for a given application to have a higher priority over the CPU to be ran 5 Interrupts in 8051 (with order of normal priority) 1. External hardware interrupts <u>INT0</u> 2. Timer 0 overflow interrupt T0 3. External hardware interrupts <u>INT1</u> 4. Timer 1 overflow interrupt T1 5. Serial interrupt (TI & RI) RESET- It is the ultimate interrupt in 8051 and is non maskable . Whenever a high level is applied to RST pin , the 8051 enters a reset condition	2+5	7	7
VII	square of a number 4300 ORG 4300H MOV DPTR, #4200 MOVX A, @DPTR MOV 0F0, A MUL AB INC DPTR MOVX @DPTR, A MOV A, 0F0 INC DPTR MOVX @DPTR, A HALT :SJMP HALT END <i>or any similar program</i> <i>(for algorithm or flowchart without program -3 mark)</i>	7*1	7	7

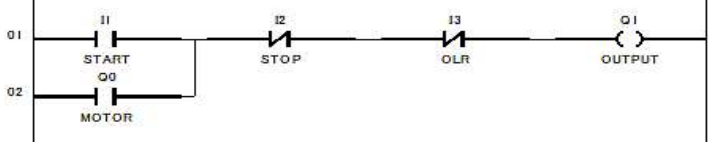
VIII	MOV R0,#20H:copy the immediate data 20H to register R0 INC R0 :Increment the content of register R0 by 1 ADD A,R5 : Add content of register R5 with content of register A and store the result in A MUL AB : Multiply the content of register A with register B ANL A,#15 h : logical AND operation between content of Register A and immediate data 15 h XCH A, R0 : exchange the contents between register A and register R0 PUSH: Increment the Stack pointer (SP) by one, and store the content of register B internal RAM location addressed by the SP.	7*1	7	7
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IX	<u>8255 Block diagram</u>  8255 is a programmable I/O device that acts as an interface between peripheral devices and the microprocessor for parallel data transfer. The three ports are Port A, Port B and Port C and as each port has 8 lines, but the 8 bits of port C is divided into 2 groups of 4-bit each. These are given as port C lower i.e., PC3 – PC0 and port C upper i.e., PC7 – PC4. And are arranged in group of 12 pins each thus designated as Group A and Group B. <i>Block Diagram:5</i> <i>Explanation:2</i>	3+4	7	7
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X	<u>Interfacing of 8051 microcontroller with DC motor .</u> <i>Sketch- 4</i> <i>Explanation - 3</i> The main purpose of DC interfacing with 8051 microcontrollers is for controlling the speed of the motor or direction of rotation  Motor driver uses the H-bridge concept. The voltage can be made to flow in either direction using this circuit (H-bridge) such that by changing the voltage direction the motor direction can be changed.  <i>Sketch- 4</i> <i>(or similar diagram)</i> <i>Explanation - 3</i> <i>(or algorithm)</i>	3+4	7	7
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XI	<u>Selection</u> of I/O's: • Type of I/O's • Memory and Programming Requirements • Compact or Modular PLC • Instruction Set/CPU • PLC Scan Time • Sinking & Sourcing PLC • Manufacturer's Support and Backup <i>(any seven point)</i>	7*1	7	7
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XII	Compare conventional and PLC based control panel (any seven point)	7*1	7	7																								
	<table><tr><td></td><td>PLC</td><td>Relay</td></tr><tr><td>1. Basic</td><td>It is a solid-state computerized industrial controller that performs software logic by using input & output modules, CPU, memory, and others.</td><td>Relay is an electro-mechanical switching hardware device (Hardware Switching Device)</td></tr><tr><td>2)Function</td><td>It plays a monitoring as well as controlling role in designing circuits.</td><td>It plays only a controlling role in the designing circuit. Monitoring is not so easy with a relay.</td></tr><tr><td>3) Working:</td><td>In PLC, we can write the program using different types of programming languages.</td><td>In Relay, we cannot write the program.</td></tr><tr><td>4) Design:</td><td>Easily modify the designing circuit.</td><td>Modification of the electronic circuit is more difficult as compared to PLC</td></tr><tr><td>5)Memory</td><td>consist of memory to store programme</td><td>no memory</td></tr><tr><td>6) I/O</td><td>It has more capabilities of input and output modules.</td><td>The relay does not have more capabilities.</td></tr><tr><td>7) Operation:</td><td>PLC is operated on the digital system.</td><td>It is operated on the analog system.</td></tr></table>		PLC	Relay	1. Basic	It is a solid-state computerized industrial controller that performs software logic by using input & output modules, CPU, memory, and others.	Relay is an electro-mechanical switching hardware device (Hardware Switching Device)	2)Function	It plays a monitoring as well as controlling role in designing circuits.	It plays only a controlling role in the designing circuit. Monitoring is not so easy with a relay.	3) Working:	In PLC, we can write the program using different types of programming languages.	In Relay, we cannot write the program.	4) Design:	Easily modify the designing circuit.	Modification of the electronic circuit is more difficult as compared to PLC	5)Memory	consist of memory to store programme	no memory	6) I/O	It has more capabilities of input and output modules.	The relay does not have more capabilities.	7) Operation:	PLC is operated on the digital system.	It is operated on the analog system.			
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XIII	 Inputs: START- To start the motor- NO STOP - To stop the motor-NC OLR- Overload Relay Output: MOTOR- Motor Sequence of Events : 1. When the Start push button is pressed, Motor has to start. 2. If the Start is released and the Stop pushbutton is not pressed, Motor should remain on. 3. When the Stop push button is pressed, Motor has to stop. 4. If stop push button is released and start is not pressed (released) motor should remain off. 5. When an overload occurs the motor stops Program - 5 Explanation-2 Or any similar programmes	5+2	7	7
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XIV	Star-delta starter ladder logic or similar Main contractor is used to supply power to the windings. It must be turned on all the time. Initially the star contactor is closed while delta contactor is open It makes the motor windings in star configuration. When the motor gains speed, the star contactor is opened while delta contactor is closed turning the motor windings into delta configuration. The contactors are controlled by using PLC. ladder logic: 4 Explanation: 3	4+3	7	7
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